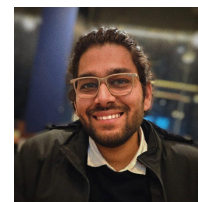


SAGAR KUMAR

Assistant Professor | Researcher | Quantum Computing, AI, Android & MEAN Stack Specialist

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RESEARCH INTERESTS

Quantum Computing • Quantum Machine Learning • Quantum Space Technology • Machine Learning • Data Science • Android App Development • MEAN Stack & Full-Stack Web Development

EDUCATION

Master of Engineering (CSE) | ITMO University, St. Petersburg, Russia
2019 – 2022

Bachelor of Technology (CSE) | ICFAI University, Jaipur
2015 – 2019

PROFESSIONAL EXPERIENCE

International Relation Coordinator & Assistant Professor | Poornima University, Jaipur
June 2025 – Present

- International Engagement & Partnerships: Coordinate global partnerships, student and faculty mobility, international collaborations, and institutional outreach to strengthen the university's global presence.
- Internationalisation & Operations: Promote internationalisation initiatives, organise global engagement activities, and manage partnership data, mobility records, and departmental operations.
- Teaching UG and PG courses in Computer Science & Engineering including Quantum Computing, AI, Android Application Development, and MEAN Stack Web Development.
- Conducting research in Quantum Computing, Quantum Machine Learning, and AI-driven systems.
- Mentoring students on research projects, academic publications, and industry-aligned software projects.

Visiting Assistant Professor | ICFAI University, Jaipur
August 2023 – June 2025

- Taught 7 courses across UG and PG levels: Machine Learning, Big Data Analytics, Data Science, Fuzzy Logic, Wireless Networks, and Computer Networks.
- Supervised student projects integrating quantum computing, AI, and full-stack development methodologies.
- Contributed to curriculum design and academic development activities.

Founder & CEO | Kosmatron, Jodhpur
2023 – Present

- Founded and led a technology startup focused on quantum computing, space-technology research, and software product development.
- Driving product development in quantum-AI, mobile application, and web platform domains.

Data Scientist | Mobyink Innovations, Jaipur
2022 – 2023

- Developed predictive analytics and machine learning models for business intelligence applications.

Data Engineer | Sense Techno Solutions, Jodhpur
2022

- Designed and maintained data pipelines and ETL workflows using AWS and cloud platforms.

COURSES TAUGHT

Undergraduate (UG): Big Data Analytics (Data Science) | Machine Learning (CSE) | Fuzzy Logic (Data Science) | Data Science (CSE) | Wireless Networks (AI & IoT) | Computer Networks (AI & IoT) | Android Application Development (CSE) | MEAN Stack Web Development (CSE)

Postgraduate (PG): Machine Learning (CSE) | Generative AI & LLMs | Quantum Computing | AI | Full-Stack Web Development (MEAN)

MTech Project Guidance: Guiding 2 Scholars for Masters in Computer Science with AI & Quantum Computing as research domains

PUBLICATIONS & BOOK REVIEWS

Research Papers & Book Chapters (Published):

- **Identification of Plant Diseases from Leaf Images Using Lightweight CNN Model** — *Q3 Journal Paper* **Published** [DOI: <https://doi.org/10.5391/IJFIS.2026.26.3.251>]
- **Optimizing K-Means Clustering with Focus on Time-Efficient Algorithms** — *Book Chapter* **Published** [doi.org/10.1201/9781003598152-16]
- **Exploring Quantum Cognition: Linking Algebraic Structures to Cognitive Phenomena** — *Book Chapter* **Published** [DOI: 10.4018/979-8-3693-4107-0.ch005]

Journal Articles & Book Chapters (Accepted):

- **Accelerated Material Discovery for Robotic Sensors Using Quantum-AI Enhanced FBG Technology** — *De Gruyter* **Accepted**
- **Regulatory and Ethical Framework for Machine Learning-driven 6G Wireless Networks** — *CRC Press* **Accepted**

Book Review:

- **Building Quantum Software** — *Manning Publications (C. Gonciulea & C. Stefanski)* **Reviewed — 2024**

PATENTS

- **Computing System for In-Orbit Simulation of Black Hole and Early-Universe Dynamics** — **Patent Published**

RESEARCH & DEVELOPMENT PROJECTS

1. SmartCampus — Android Academic Management Application [[Android](#) | [Kotlin](#) | [Firebase](#)]

- Designed and developed a full-featured Android application in Kotlin to digitise campus operations for students, faculty, and administrators on a single unified platform.
- Implemented real-time features — attendance tracking, timetable management, grade notifications, and campus announcements — using Firebase Realtime Database and Firebase Cloud Messaging (FCM).
- Built a clean Material Design 3 UI with XML Layouts and Jetpack components: ViewModel, LiveData, Navigation Component, and Room Database for offline-first functionality.
- Integrated Google Maps SDK for campus navigation, Retrofit for REST API consumption, and OAuth 2.0 for secure multi-role authentication.
- Conducted user-acceptance testing with 200+ student participants; iterated on feedback to improve UX, achieving a 4.6/5 average satisfaction score.
- Tech Stack: Kotlin, Android Studio, Firebase (Auth, Firestore, Realtime DB, FCM, Storage), Jetpack, Retrofit, Room DB, Google Maps SDK, Material Design 3.

2. AcademiQ — MEAN Stack Learning Management Portal [[MongoDB](#) | [Express.js](#) | [Angular 17](#) | [Node.js](#)]

- Architected and built a full-stack Learning Management System (LMS) enabling faculty to publish course content and students to access assignments, quizzes, and live sessions through a browser-based SPA.
- Developed a RESTful API backend with Node.js and Express.js featuring JWT-based authentication, role-based access control (RBAC), input validation, and rate limiting.

- Used MongoDB Atlas with Mongoose ODM for flexible schema design; implemented aggregation pipelines for real-time analytics dashboards showing enrolment trends, submission rates, and grade distributions.
- Built the frontend in Angular 17 with NgRx state management, RxJS reactive streams, lazy-loaded modules, Angular Material components, and reactive forms with server-side validation.
- Added real-time notifications and live class indicators via Socket.io; integrated Cloudinary for media uploads and SendGrid for automated email reminders.
- Deployed on AWS EC2 behind NGINX reverse proxy with PM2 process management and set up GitHub Actions CI/CD pipeline for automated testing and zero-downtime deployment.
- Tech Stack: MongoDB Atlas, Mongoose, Express.js, Angular 17, NgRx, Node.js, TypeScript, Socket.io, JWT, AWS EC2, NGINX, PM2, Docker, GitHub Actions, Cloudinary.

3. Stock Market Prediction LLM & Finance Management Platform [\[Python | LLMs | NLP | REST APIs\]](#)

- Developed an end-to-end AI platform integrating Large Language Models with quantitative finance algorithms for real-time stock market prediction and portfolio management.
- Engineered NLP pipelines to extract sentiment and financial signals from news feeds, earnings reports, and social media streams using transformer-based models.
- Built a finance management dashboard with automated risk assessment, portfolio optimisation, and predictive analytics; deployed on cloud-native architecture with RESTful APIs.

4. Quantum_Speed_Up_NLP — Hybrid Quantum-Classical NLP Model [\[Qiskit | PennyLane | PyTorch | BERT\]](#)

- Led development of a hybrid NLP pipeline integrating quantum computing with classical deep learning for sentiment analysis on the IMDB dataset.
- Designed parameterised quantum circuit layers in PennyLane and Qiskit, integrated seamlessly with PyTorch for end-to-end gradient-based training.
- Benchmarked hybrid model against classical BERT baselines; documented performance improvements and computational trade-offs.

5. Quantum Supervised Learning for Healthcare AI — Hybrid Disease Prediction [\[PennyLane | PyTorch | Scikit-learn\]](#)

- Designed a hybrid ML pipeline combining classical neural networks with quantum circuit-based feature extraction using PennyLane.
- Developed custom quantum layers to enrich input features for disease prediction on the Pima Indians Diabetes Dataset, improving classification accuracy over classical baselines.
- Validated quantum-enhanced features in a practical healthcare setting, demonstrating measurable benefits of quantum ML on real-world medical data.

6. Neo-Nash: Quantum Nash Equilibrium Solver [\[Qiskit | VQE | Game Theory\]](#)

- Implemented a Variational Quantum Eigensolver (VQE) to approximate Nash equilibria in multiplayer, multi-strategy game theory problems.
- Engineered modular classes for payoff matrix validation and Hamiltonian encoding; designed parameterised ansatz circuits with custom cost functions.
- Demonstrated quantum advantage over classical solvers on computationally hard game scenarios.

7. Neutrino Oscillation Simulation [\[Python | NumPy | Matplotlib | Streamlit\]](#)

- Designed a Python simulation framework to model neutrino oscillation probabilities in vacuum and matter using NumPy, Pandas, and Matplotlib.
- Supported flexible parameter tuning with automated plot generation and CSV/JSON export for experimental validation.
- Proposed extensions for 4-flavour oscillations, live data integration, and a Streamlit GUI.

8. Quantum-Enhanced Protein Folding & Structural Analysis Engine [\[VQE | Quantum Annealing | Docker | Kubernetes\]](#)

- Built a hybrid platform combining VQE-based quantum simulation and deep neural networks to predict protein folding pathways and identify druggable sites.
- Engineered noise mitigation strategies for higher-fidelity energy landscape exploration; containerised with Docker and deployed on Kubernetes for scalable cloud execution.

9. Personalized Medicine via Quantum-AI Biomarker Discovery [\[Quantum PCA | Quantum Clustering | Deep Learning\]](#)

- Developed a platform combining Quantum PCA and quantum clustering to mine subtle patterns in large-scale genomic, proteomic, and metabolomic datasets.
- Designed a personalised treatment recommendation engine linking biomarker profiles to individual patient data for customised therapy planning.

CREDLY BADGES AND SKILLS

(Profile: <https://www.credly.com/users/sagar-kumar.a51d16f1>)

- **Basics of Quantum Information** — *By IBM*
[https://www.credly.com/badges/c230fc79-9c89-4979-954e-8a1a671b8cba/public_url]
- **IBM Quantum Challenge 2024 Achievement** — *By IBM*
[https://www.credly.com/badges/0abe1136-32ff-462b-a3c6-62e61b523a7b/public_url]
- **Practical Introduction to Quantum-Safe Cryptography** — *By IBM*
[https://www.credly.com/badges/72f9df31-2d12-47ee-9759-d60f64e9e74d/public_url]
- **Qiskit Global Summer School 2024 - Quantum Excellence** — *By IBM*
[https://www.credly.com/badges/7b997aa1-cd12-4f85-b19f-7f1627c9a127/public_url]
- **Variational Algorithm Design** — *By IBM*
[https://www.credly.com/badges/8e23e93d-df90-4f4c-a7d7-415b0b9d5fc0/public_url]
- **LFQ101: Fundamentals of Quantum Computing** — *The Linux Foundation*
[https://www.credly.com/badges/733fe48c-4f14-4510-9be3-10e37f9f16ec/public_url]
- **LFQ102: Quantum Computing Essentials For Senior Leaders** — *The Linux Foundation*
[https://www.credly.com/badges/6bd08c72-91d4-4d06-be3c-ae76b9d981e2/public_url]

FACULTY DEVELOPMENT PROGRAMS

- Harnessing Artificial Intelligence for Innovative Teaching and Learning in Higher Education — Research & Development Cell, Poornima University
- NEP 2020 Orientation and Sensitization Programme under Malaviya Mission Teacher Training Programme (MM-TTP), University Grants Commission — UGC Centre, Maharshi Dayanand Saraswati University, Ajmer

CERTIFICATIONS

Quantum & AI

- Quantum Introduction — QubitXQubit & IBM
- Quantum Challenge — IBM
- Machine Learning — Ravindra Babu Ravula
- Elements of AI — University of Helsinki

Android Development

- Android App Development with Kotlin — JetBrains Academy / Google Developer Training
- Android Development Bootcamp (Kotlin & Jetpack) — Udemy
- Firebase for Android Developers — Google Codelabs

MEAN Stack & Web Development

- Node.js, Express, MongoDB & More: The Complete Bootcamp — Udemy
- Angular — The Complete Guide (2024 Edition) — Udemy
- Complete MEAN Stack Web Developer Bootcamp — Udemy
- TypeScript for Professionals — Udemy

Databases & Cloud

- Introduction to MySQL — IIT Bombay
- Human Centred Robotics — IIT Jodhpur & Robotics Society of India

TECHNICAL SKILLS

Programming Languages

Core Languages: Python, JavaScript, TypeScript, Kotlin, Java, MATLAB, SQL (MySQL), HTML5, CSS3

Android Development

Languages & IDE: Kotlin, Java, Android Studio, XML Layouts

Jetpack Components: ViewModel, LiveData, Navigation, Room Database, WorkManager, DataStore

Networking & APIs: Retrofit, OkHttp, RESTful APIs, OAuth 2.0, JSON parsing

Backend Services: Firebase (Auth, Firestore, Realtime DB, FCM, Cloud Storage), Google Maps SDK

UI & Design: Material Design 3, ConstraintLayout, RecyclerView, ViewBinding, Glide / Coil

Testing & Publishing: JUnit, Espresso, APK Signing, Build Variants, ProGuard, Google Play Store

MEAN Stack & Web Development

MongoDB: MongoDB Atlas, Mongoose ODM, Aggregation Pipelines, Indexing & Query Optimisation

Express.js: RESTful API Design, Middleware, JWT Authentication, Role-Based Access Control (RBAC), Error Handling

Angular: Angular 17, NgRx State Management, RxJS, Angular Material, Lazy Loading, Reactive Forms, SSR

Node.js: Event Loop, Async/Await, Socket.io (real-time), PM2, NGINX Reverse Proxy, Rate Limiting

DevOps: Docker, Kubernetes, GitHub Actions CI/CD, AWS EC2, NGINX, Cloudinary, SendGrid

Quantum Computing & AI / ML

Quantum Frameworks: Qiskit, PennyLane, Cirq

AI / ML Libraries: PyTorch, TensorFlow, Scikit-learn, BERT, HuggingFace Transformers, LangChain

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly

REFERENCES

Name	Position	Email	Phone
Dr. Rajesh Ranjan	Asst. Prof. @ LNMIT	rajesh.ranjan@lnmiit.ac.in	+91-8890003228
Dr. Pranay Bhardwaj	PDF @ New York Univ.	pranaybhardwaj@nyu.edu	+91-9001247742

DECLARATION

I hereby solemnly declare that all information furnished in this curriculum vitae is true and correct to the best of my knowledge and belief. If at any time I am found to have concealed or misrepresented any material information, my appointment shall be liable to termination without notice or compensation.

Place: Jaipur **Date:** 05/07/2025

Sagar Kumar